Práctica tres: Modelado y control del sistema musculoesquelético

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Información general



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Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('sistema3')
parameters.StopTime = '10'
```

```
parameters =
struct with fields:
    StopTime: '10'
```

```
parameters.Solver = 'ode15s'
```

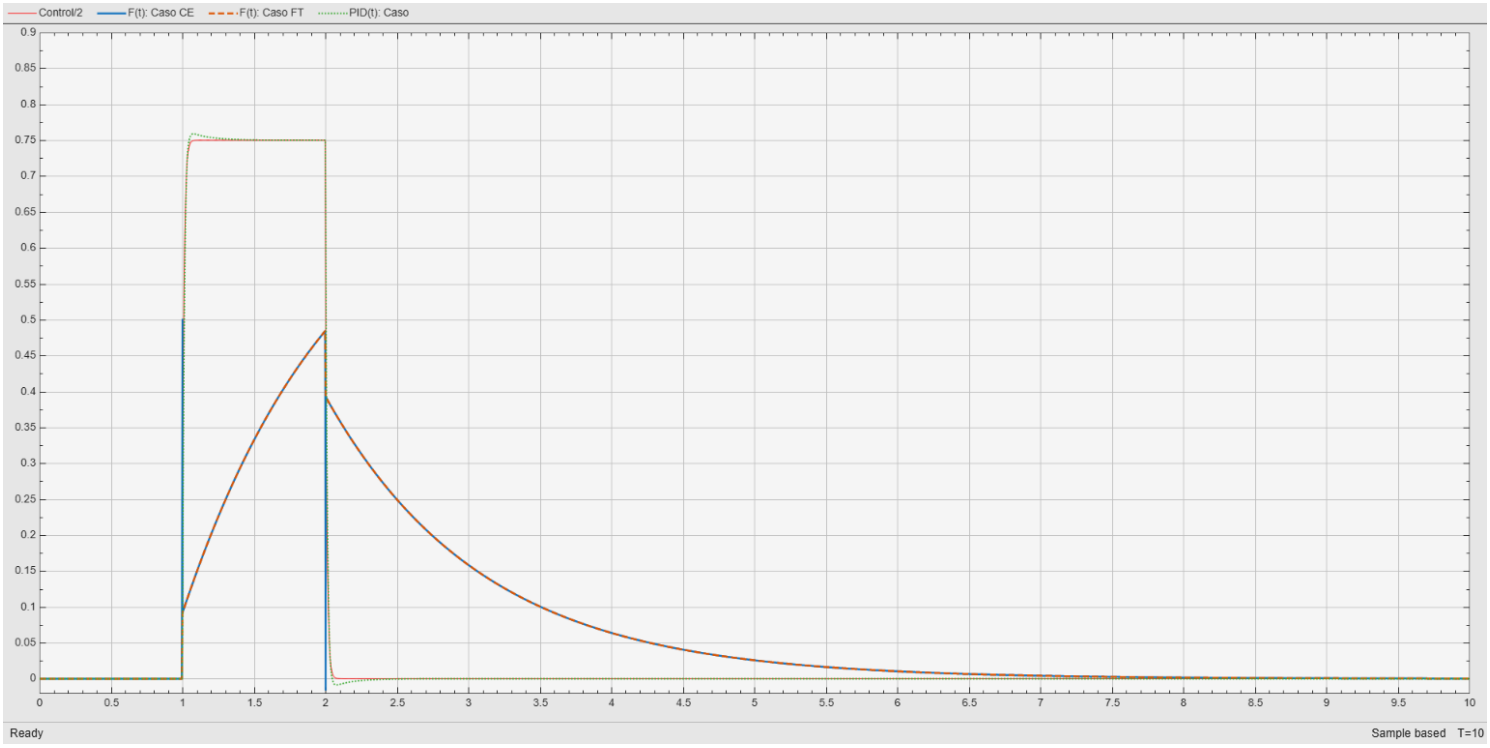
```
parameters =
struct with fields:
    StopTime: '10'
      Solver: 'ode15s'
```

```
parameters.MaxStep = '1e-3'
```

```
parameters =
struct with fields:
    StopTime: '10'
      Solver: 'ode15s'
     MaxStep: '1e-3'
```

Respuesta: Impulso Unitario

```
x=sim('sistema3',parameters)
```

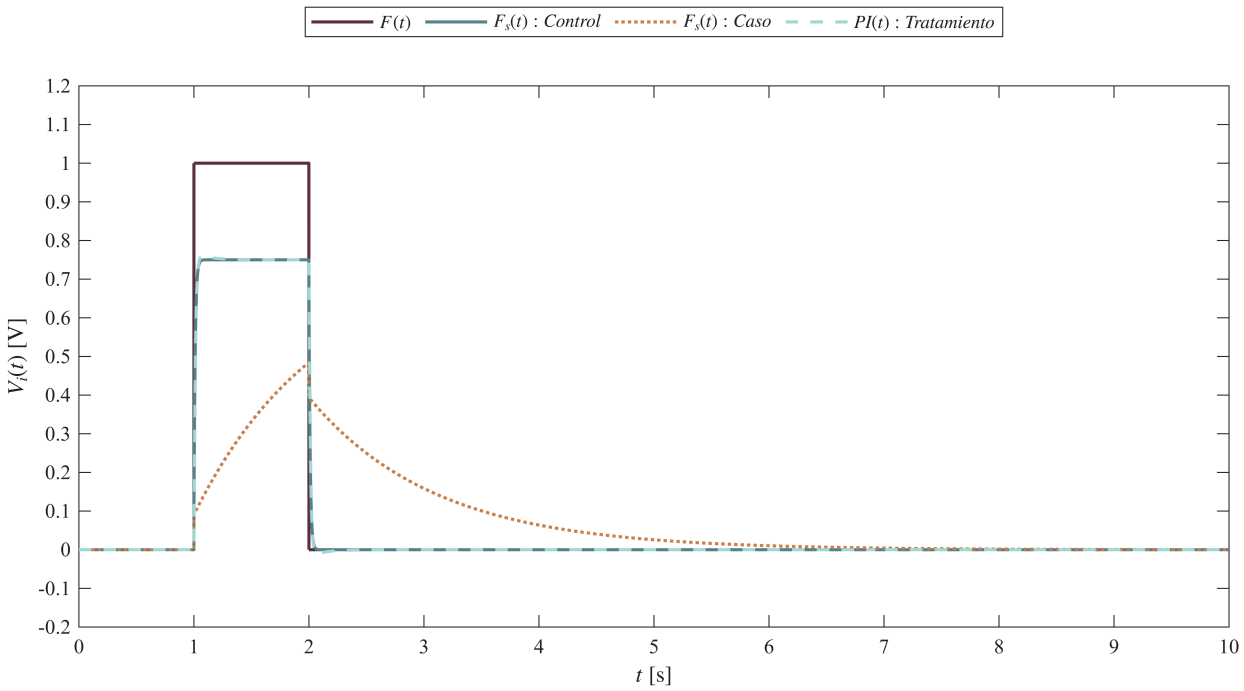



```
x =
  Simulink.SimulationOutput:

                    F: [10124x1 double]
                  Fsx: [10124x1 double]
                  Fsy: [10124x1 double]
                  PID: [10124x1 double]
                    t: [10124x1 double]

    SimulationMetadata: [1x1 Simulink.SimulationMetadata]
          ErrorMessage: [0x0 char]
```

```
plotsignals(x.t,x.F,x.Fsx,x.Fsy,x.PID,'Caso')
```



Funcion: Respuestas de la simulacion

```
function plotsignals(t,F,Fsx,Fsy,PID,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position', [1,1,10,5])
    set(gca, 'fontname','times new roman','fontsize',10)
    hold on; grid off; box on;
    clr1 = [92, 46, 60]/255;
    clr2 = [87, 128, 136]/255;
    clr3 = [202, 126, 76]/255;
    clr4 = [157, 217, 210]/255;
    order = [clr1;clr2;clr3;clr4]; colororder(order)


plot(t,F,'-',t,Fsx,'-',t,Fsy,':',t,PID,'--','LineWidth',1.5,'MarkerSize',5,'MarkerIn
dices',...
        1:1e3:length(t));

    L = legend('$F(t)$','$F_{s}(t): Control$','$F_s(t): Caso$','$PI(t):
Tratamiento$');

set(L,'interpreter','latex','location','northoutside','orientation','horizontal')

    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$V_i(t)$ [V]', 'Interpreter','latex')

    xlim([0 10]); xticks(0:1:10)
    ylim([-0.2 1.2]); yticks(-0.2:0.1:1.2)
```

```
    exportgraphics(gcf,[signal, '_matlab.pdf'],'contenttype','vector')

end
```